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NMAP to Discover Live Hosts using Nmap Scan (ARP, ICMP, TCP/UDP) on the TryHackMe Platform

Aim:

To discover live hosts using Nmap scans such as ARP, ICMP, TCP & UDP scans on the TryHackMe platform.

Introduction:

Nmap (Network Mapper) is a network scanning & host discovery tool. it is used to identify active systems in a network & gather information about the available hosts and services. Nmap can use different techniques to discover whether a system is active. common host discovery techniques include ARP scan, ICMP scan & TCP/UDP ping scan.

Host Discovery Methods:

i. ARP Scan:

Arp (Address Resolution Protocol) is mainly used to discover host in the local network. An ARP scan sends ARP requests to the same local network.

ii. ICMP Scan:

ICMP scan is commonly used to check whether a host is reachable. An ICMP scan sends ICMP requests to the target host &

checks for a response. If the host responds it can be identified as a live host.

iii) TCP/UDP Scan:

TCP/UDP packets can also be used for host discovery. Nmap sends suitable TCP or UDP packets to the target and analyses the response to determine whether the host is active.

Procedure:

* Open the TryHackMe Nmap environment.
* Identify the target IP address or IP address range to be scanned.
* Start the Nmap host discovery process.
* Perform an ICMP scan to check the reachability of the target hosts.
* Perform TCP/UDP ping scans to discover hosts that respond to TCP or UDP packets.
* Observe the responses received from the target systems.
* Identify the hosts that are currently active or live.
* Record the discovered IP addresses and scan results.
* Compare the results obtained using the different scanning methods.

Result:

Thus the live hosts in the target network were successfully discovered using Nmap ARP, ICMP & TCP/UDP host discovery scans.